



Water Meter Reading Recognition Based on Lightweight CNN

Fumin Zou

Fujian Key Laboratory of Automotive
Electronics and Electric Drive, Fujian
University of Technology
fmzou@fjut.edu.cn

Zhimeng Shi

Fujian Key Laboratory of Automotive
Electronics and Electric Drive, Fujian
University of Technology
1192282696@qq.com

Zhenhua Gan*

Fujian Key Laboratory of Automotive
Electronics and Electric Drive, Fujian
University of Technology
ganzh@fjut.edu.cn

Lvchao Liao

Fujian Key Laboratory of Automotive
Electronics and Electric Drive, Fujian
University of Technology
achao@fjut.edu.cn

Jiamin Xu

Fujian Key Laboratory of Automotive
Electronics and Electric Drive, Fujian
University of Technology
827938160@qq.com

ABSTRACT

In order to improve the accuracy of reading recognition in water meter image, a reading recognition method based on lightweight convolution neural network is proposed. The water meter image collected needs to be preprocessed by the methods of filtering, Hough circle detection, grayscale, edge detection, character area positioning and tilt correction. Then, a lightweight convolutional neural network consists of five convolution layers is proposed, which was confined to process the limited area of numeric characters in the water meter images. In addition, a certain number of 3×3 , 1×3 and 3×1 convolutional layers were reconstructed from 3×3 and 5×5 structures to simplify the parameters and complexities of the model. In addition to improving the recognition accuracy, the features before asymmetric convolution were fused with that after asymmetric convolution. The experimental result shows that the training parameters are reduced greatly and the efficiency of training is improved significantly. And under the 3000 iterations, the recognition accuracy of this improved network is 98.85%, while the recognition accuracy of LeNet-5 network is only 97.11%.

CCS CONCEPTS

• **Computing methodologies** → Machine learning; Machine learning approaches; Neural networks; Computer graphics.

KEYWORDS

Image preprocessing, Lightweight model, Convolutional neural network, Feature fusion, Character recognition

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1 INTRODUCTION

Automatic meter reading (AMR) is an indispensable part of the process of smart city construction. At present, there are still a large number of traditional mechanical water meters without the function of the digital data transmission. How to intelligentize these water meters is a great challenge. Normally, installing a micro camera above the water meter and recognizing the reading in the collected water meter image is a common way to reform the water meter [1]. However, the information of the collected water meter images is quite varied because of environmental conditions. In order to eliminate the irrelevant information and simplify the data in the images as much as possible, the images should be grayed and filtered first. It can contribute to the reliability of image segmentation and feature extraction. Subsequently, the reading of the water meter can be obtained by character recognition technology [2]. Up to now, the main methods of character recognition have included template matching, feature matching and neural network [3]. Kusumadewi, Ira, Christy Atika Sari et al. [4] used a method that combines bounding box and template matching to separate and recognize each character. This method has the advantages of fast calculation speed, but the recognition accuracy is not high enough. Wen, Ying et al. [5] adopted a way of character recognition based on support vector machine, which has strong robustness to the changes of illumination and color of image in complex environment. But it is not effective for character recognition in low resolution images. Yi, Chucai, Xiaodong Yang et al. [6] proposed the GHOG based on HOG, which adopts global sampling. It can build a better model of character structure and improve the accuracy of character recognition. However, the efficiency and stability of this method need to be further improved.

At the same time, with the development of deep learning, a variety of character recognition algorithms based on convolutional neural network (CNN) have been proposed by lots of researchers [7]. CNN has the characteristics of local sensing, weight sharing and downsampling and it can train CNN model with less computing resources and less time. Especially, the great self-learning ability and recognition generalization ability are its main advantages [8]. However, in order to improve the computational performance of

CNN, it is generally necessary to increase the number of layers of the network [9]. Even it is required to use more expensive image workstations with higher computing performance so that more complex algorithms can be run. The methods of character recognition introduced in the front of this paper, which can recognize numbers, letters and even text, so they are widely used in license plate recognition and text character recognition. In contrast, the reading of the water meter is only related to the numeric characters [10], and the features in the image have been reduced a lot after graying and filtering, so it is not necessary to build too complex neural network model for this work. On the basis of the existing research, in order to further improve the recognition accuracy of water meter reading, this paper proposes a lightweight CNN model. It improves the original neural network by splicing a certain number of convolution layers of 3×3 , 1×3 and 3×1 . Furthermore, the features before asymmetric convolution were fused with that after asymmetric convolution. This method not only improves the character recognition ability, but also simplifies the parameters and complexity of the model.

2 PROPOSED METHODS

This section will introduce the key methods of water meter reading recognition. Firstly, the method of cutting characters in water meter image is introduced, and then the construction method of the lightweight CNN is given.

2.1 Image Preprocessing

The installation environment of water meters are often complex and changeable. It is easy to be affected by various factors when collecting the images. The problems of low contrast, image blur and noise will affect the accuracy of character recognition. Therefore, the images must be pre-processed so that the digital area of water meter will be located accurately and the accuracy of water meter reading recognition will be improved.

2.1.1 Graying The Image. The color features of the image can be discarded to reduce the amount of original image data. The RGB color image should be converted into a 8-bit grayscale image firstly [11]. The methods of component, maximum value and weighted average are commonly used in image grayization. By contrast, it is planned to use the method of weighted average to gray-scale the image. The calculation formula is as follows:

$$g(x, y) = 0.9f(x, y)_R + 1.77f(x, y)_G + 0.33f(x, y)_B \quad (1)$$

2.1.2 Image Filtering. The image should be filtered to further suppress the noise in the target image. The methods of mean filtering, median filtering and Gaussian filtering are the most common in image filtering [12]. Owing to Gaussian noise accounts for a large proportion in the image noise, the method of Gaussian filtering is adopted in this paper. The function of two-dimensional Gaussian is as follows:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad (2)$$

2.1.3 Character Region Location. The steps and results of locating the character area are shown in Figure 1. Hough circle detection [13] is used to extract the circular dial in the image after filtering, so that the dial and the background can be separated. The Canny

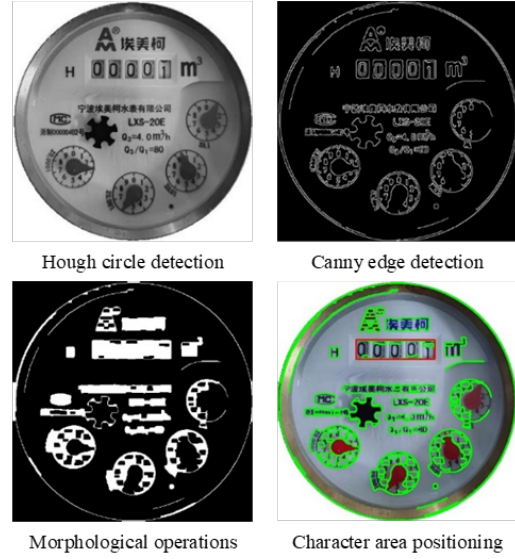


Figure 1: Character region location

edge detection algorithm [14] can be used to mark the areas with edges due to the sharp edges of the digital area. Subsequently, the algorithm of mathematical morphology [15] is used to fill the area. This paper mainly uses the combination of expansion and erosion. By adjusting the size of the structural elements, a more ideal connected area can be obtained. It is our goal to get the outline of the character area and locate it, but the whole dial will also be projected when the contour of the digital area is projected. So there will be some interference of the contours such as the name plate, pointer, etc. In order to find the position of the character area faster, each contour is filtered according to the area size of the circumscribed rectangles of the contours, and then the target area is found by combining with the geometric characteristics of the reading area.

2.1.4 Tilt Correction. The captured image may be tilted because of the camera angle. So it is necessary to correct the tilted character region, which is beneficial to the subsequent character segmentation. The minimum bounding rectangle covering the contour of the target region can be obtained after locating the character region. At this time, we can also get the location information of the rectangular box, such as the center coordinates, tilt angle, length and width. Because the character region is our concern, the center of the rectangular box containing the character area can be taken as the rotation center, and the tilt angle of the rectangular box can be taken as the rotation angle. Then the image can be tilt corrected by affine transformation. The images before and after tilt correction are shown in Figure 6 and Figure 7

2.1.5 Character Segmentation. We can capture the image of the character area according to the position of the rectangle after correcting the image skew, as shown in Figure 4. The recognition effect of taking the string as the unit directly is not good enough. In contrast, taking a single character as the unit is a common character recognition method. Therefore, it is necessary to divide the



Figure 2: Before correction



Figure 3: After correction

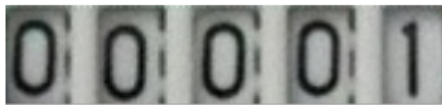


Figure 4: Water meter character area

character region into a single character. The accuracy of character recognition will be affected by the accuracy of single character segmentation directly.

Firstly, the character images are processed by graying and binarization. Furthermore, the binarization image can be expanded



Figure 5: Expanded binary image

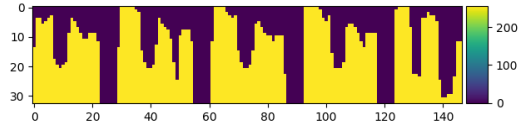


Figure 6: Vertical projection



Figure 7: Single character after segmentation



Figure 8: Character normalization

in the vertical direction to facilitate the cutting of characters, as shown in Figure 5

Then, the vertical projection division method is used to count the white pixels in the vertical direction of each column in the character image from left to right. The statistical results are shown in Figure 6. According to the peaks and troughs in the projection image, each character can be divided in the vertical direction. Similarly, the projection in the horizontal direction can also be cut by this method. The final segmentation effect is shown in Figure 7

After each character is segmented, it is necessary to normalize the single character image. Finally, the character images with the size of 28×28 are obtained, as shown in Figure 8. It is in preparation for subsequent character recognition.

2.2 Lightweight Convolutional Neural Network

The improved lightweight CNN model is shown in Figure 9. The character image is taken as the input layer after preprocessing and segmentation, and the dimension of the image is $28 \times 28 \times 1$. The whole network structure consists of five convolution layers, and convolution kernels with sizes of 3×3 , 1×3 and 3×1 are used to extract image features respectively. The sliding step of convolution kernel is set to 1, and the ReLU activation function is used after each convolution layer. The maximum pool schemes with filter size of 2×2 are used in the pooling layers. At the same time, a dropout ratio of 0.25 is added to improve the generalization ability of the model.

Referring to the network structure of LeNet-5[16], this paper replaces the first convolutional layer of the LeNet-5 network with two convolutional cores of 3×3 , as shown in Figure 10. The depth of the network and the number of activation functions are increased by this method. It is conducive to improve the nonlinear characteristics

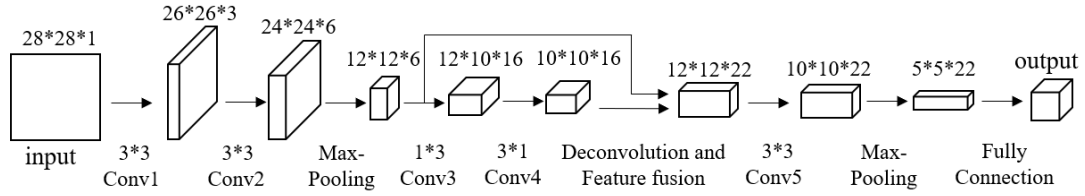


Figure 9: Improved network model

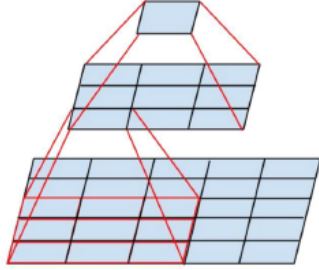


Figure 10: First replacement

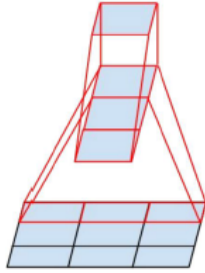


Figure 11: Second replacement

of the network, and the training parameters are reduced effectively while keeping the receptive field size unchanged. On the whole, the efficiency of calculation is improved. In addition, Szegedy C et al. [17] also pointed out that the $n \times n$ convolution kernel can be replaced by the asymmetric convolution kernel when the size of the feature graph is between 12 and 20.

It can further reduce the training parameters and ensure the performance of the network. As shown in Figure 11, after the first pooling, the 3×3 convolution layers are replaced by 1×3 and 3×1 convolution layers, and the number of convolution cores in each layer is 16. The global information of each pixel in the convolution generated feature map is reduced and the local details are increased after replacement. In this way, more digital shape features can be obtained, which helps to improve the recognition accuracy.

In the process of adjusting the network structure, it was found that the errors of image recognition are mainly caused by the small size of characters and the similarity of some structures between numbers. Multi-scale feature fusion [18] is often used in target detection network to improve the accuracy of small target detection. In this paper, the combined features before and after convolution are fused in series, and then the fused feature map is convoluted



Figure 12: General images



Figure 13: Complex images

and pooled again. Finally, there is a full connection layer which contains 550 neurons and passes through the activation function of ReLu. Besides, the dropout in this network is set to 0.25, and the output layer is 0-9 for each digit category.

3 EXPERIMENTAL RESULTS AND ANALYSIS

3.1 Data Preparation

Firstly, the collected water meter images were divided into general images and complex images. The characteristics of general images are that the character area is clear, the image brightness is uniform, the tilt angle is small, and the recognition difficulty is low, as shown in Figure 12. The complex images are characterized by low resolution, adhesion and large brightness difference, which are difficult to recognize, as shown in Figure 13. The two kinds of images were used to verify the accuracy of water meter character recognition in different environments.

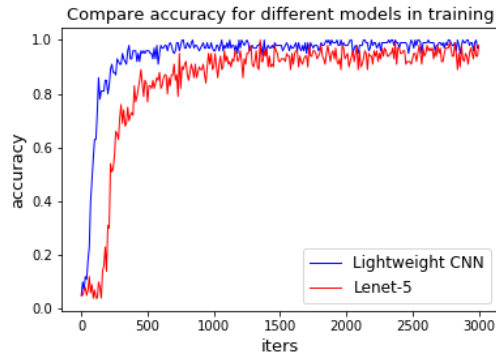
In this experiment, each number from 0 to 9 has 800 corresponding images with the size of 28×28 . The images were divided into training set and test set according to the ratio of 5:1. The multi-class log loss function was used in training, and the learning rate of the network model was set to 0.001. The algorithm of Adam was used to optimize the training parameters, and the number of training iterations was set to 3000.

3.2 Result Analysis

Since it is very important to get a single character accurately in character recognition, we need to verify the accuracy of character

Table 1: Character segmentation test results

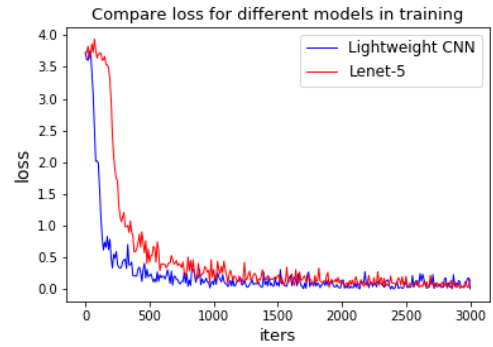
Picture type	Number of tests	Correct number	Correct rate
General image	486	471	96.91%
Complex image	241	209	86.72%

**Figure 14: Recognition accuracy**

segmentation for different images. Therefore, we use the vertical projection method to segment the characters in ordinary image and complex image respectively. The test results of character segmentation for general images and complex images are shown in Table 1. As can be seen from the table, the accuracy of general image segmentation is higher than that of complex image segmentation significantly. But on the whole, the method used in this paper has a good performance of character location and segmentation. It can process water meter image in complex environments to a certain extent.

The character recognition performance of the lightweight CNN model should be tested after obtaining the single character. Comparing the recognition results of LeNet-5, AlexNet, VGG-16[19] and our network for single character image, as shown in Table 2

Besides, the performance of the network is also evaluated by the convergence speed of the loss function and the accuracy of character recognition in the training process. As can be seen from Figure 14 and Figure 15, comparing with LeNet-5 network, the recognition accuracy of lightweight CNN is higher, and the convergence speed of its loss function is faster.

**Figure 15: Loss function**

4 CONCLUSION

In this paper, the problem of water meter recognition is solved effectively by our new network model. Aiming at improving the reliability of water meter reading recognition, a reading recognition method based on lightweight convolution neural network is proposed. The experimental results show that this method can recognize the pre-processed image accurately. Comparing with the classical neural network, the network requires fewer parameters and less computation while identifying the reading of water meter accurately. In the following work, we will further optimize our model architecture and algorithm from other perspectives, e.g., running our algorithm on smart cameras.

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Table 2: Character recognition test results

Model	Number of correct identification	Number of misidentifications
Lightweight CNN	2328	27
LeNet-5	2287	68
AlexNet	2299	56
VGG-16	2307	48

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